MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT S-541( )/TTC-38(V)

This specification is approved for use by all Departments and Agencies of the DOD.

1. SCOPE

1.1 Scope.- This specification covers the air and vehicular transportable shelter facility part of Control-Test-Maintenance Group OK-267( )(V)/TTC-38(V) which is used with the Central Office, Telephone, Automatic AN/TTC-38. (See 6.1 and 6.5)

1.2 Description.- Shelter, Electrical Equipment, S-541( )/TTC-38(V), hereinafter called the shelter facility, includes a modified S-280( )/G shelter with work benches, racks, brackets, storage drawers and shelves, tie-downs, electrical components and wiring to enable installation and operation of the assemblage components of the OK-267( )(V)/TTC-38(V). (See 6.1 and 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)

FSC 5410

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MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys & Titanium, Spot & Seam
MIL-T-7928	Terminals, Lug & Splice, Crimp Style Cooper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-W-45205	Welding, Gas, Metal-Arc & Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-P-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging & Packing of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures & Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual & Mechanical Defects
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1235	Single and Multi-level Continuous Sampling Procedures & Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods
MIL-STD-1474	Noise Limits for Army Material

DRAWINGS

ELECTRONICS COMMAND

SC-D-629640	Control Test Maintenance Group OK-267(V)1/TTC-38(V)
DL-SC-B-629641	Shelter, Electrical Equipment S-541( )/TTC-38(V)
SC-D-629721	Power Schematic S-541( )/TTC-38(V)
SC-D-629722	Signal Schematic S-541( )/TTC-38(V)

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First Article.- When specified in the contract, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.- The shelter facility shall be constructed in accordance with drawings listed on DL-SC-B-629641 and the requirements specified herein. (See 4.6)

3.3 Electrical Installation.- (See 3.23)

3.3.1 Power system wiring and components.- The completed in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of conductors.- Continuity shall exist between each pin of the power entrance connectors and respective circuit terminations and shall be less than 1 ohm. (See 4.7.1 and 4.7.1.1)

3.3.1.2 Dielectric breakdown voltage.- The insulation of all power wires, cables, and components shall withstand, without flashover or insulation breakdown, a potential of 500 volts DC applied continuously for not less than 30 seconds. (See 4.7.1 and 4.7.1.2)

3.3.1.3 Insulation resistance.- The resistance of insulation between any two circuit conductors and also between each of the four ungrounded power line conductors and shelter facility ground terminal shall not be less than 100 megohms. (See 4.7.1 and 4.7.1.3)

3.3.2 Power distributional and operational functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as shown in the power schematic drawing SC-D-629721 when tested in accordance with 4.7.2 and 4.7.2.1.

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3.3.3 Signal (audio) system wiring and components.- The completed in-place signal system installation shall meet the following requirements:

3.3.3.1 Continuity.- Continuity shall exist between end terminations of all signal wires and shall be less than 5 ohms. (See 4.7.3 and 4.7.3.1)

3.3.3.2 Dielectric and insulation resistance.- The insulation of all signal wires, cables, and components shall withstand, without flashover or insulation breakdown, a potential of 500 volts DC applied continuously for not less than 5 seconds. The insulation resistance shall not be less than 100 megohms. (See 4.7.3.2)

3.3.4 Cable assemblies, power and audio frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the continuity, dielectric, insulation resistance, and pull-out requirements of MIL-C-3885. (See 4.4)

3.3.5 Suppression of electromagnetic interference.- The motors in the ventilating equipment used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings. (See 4.4)

3.4 Systems operation tests.- With all test and operational equipment of the OK-267(V)/TTC-38(V) assemblage installed, the operational requirements of all power and signal frequency cable assemblies, switches, connectors, and test equipment shall be met when tested in accordance with 4.8.

3.5 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicle without sustaining any buckling, cracking, delamination, or other permanent deformation. (See 4.9)

3.6 Rail transportation.- The shelter facility shall be capable of being loaded, blocked and braced on a flatcar without any extra tying, taping, etc., without sustaining any buckling, cracking, delamination, or other permanent deformation. (See 4.10)

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3.7 Drop tests.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete without sustaining any buckling, cracking, delamination, or other permanent deformation. (See 4.11)

3.8 Watertightness.- There shall be no evidence of leakage into the shelter facility or its walls, floor, roof or door, during or after the test. (See 4.12)

3.9 Fording.- There shall be no evidence of leakage into the shelter facility or its walls, floor or door, during or after the test. (See 4.13)

3.10 Humidity.- Hardware sample shall have no evidence of corrosion, rust or deterioration after testing. (See 4.14)

3.11 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive, and when installed, each insert shall be capable of withstanding the specified tensile and torque load. There shall be no damage or permanent distortion to the shelter facility mounting surface or insert head or threads and the insert shall not have rotated from its original position. Insert threads shall be free of paint. (See 4.15)

3.12 Acoustical Noise.- The acoustical noise shall not exceed the NC-60 criteria specified in MIL-STD-1474, Table 2, Category F at the operators, supervisors and maintenance position in the completed shelter facility. (See 4.16)

3.13 Cleaning.- (See 3.23)

3.13.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

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3.13.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware and other foreign material.

3.14 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.15 Marking.- (See 3.23 and 4.4)

3.15.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings.

3.15.2 Nameplates.- Each shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.15.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the centers-of-gravity shall be verified by the contractor using the first article.

3.15.4 Interior and exterior marking.- All interior and exterior marking of the shelter facility shall be as depicted on the drawings.

3.16 Riveting.- (See 3.23) Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.17 Seal, vapor.- (See 3.23) All joints and raw edges of the framing for the power and signal entrance boxes, air-conditioner, and exhaust blower shall be sealed to prevent entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be one part Silicone Construction Sealant Silastic 738 as made by Dow Corning, or equal, using the manufacturer recommended primer.

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3.18 Securing of parts.- (See 3.23) Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.18.1 Screws, threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.19 Soldering.- (See 3.23)

3.19.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.19.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.19.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, over-heated or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.20 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finish weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum.

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Welds shall conform to MIL-W-6858 Class B and MIL-W-45205 Class B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 3.23 and 4.4)

3.21 Wiring and cabling.- (See 3.23) Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.21.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.21.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.21.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.21.4 Solderless Terminal Lug Connections.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements of MIL-T-7928. (See 4.17)

3.21.5 Identification of wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.

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3.22 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.18) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.23 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs: (See 4.19)

3.2 Construction

3.3 Electrical installation

3.11 Mounting inserts

3.13 Cleaning

3.14 Finish

3.15 Marking

3.16 Riveting

3.17 Seal, vapor

3.18 Securing of parts

3.19 Soldering

3.20 Welding

3.21 Wiring and cabling

3.22 Interchangeability

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the

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contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform such inspection as deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.20)

4.3 First Article.- Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor. (See 3.1)

4.3.1 First Article units.- The contractor shall furnish one (1) first article unit of the complete S-541( )/TTC-38(V).

4.3.2 First Article inspection.- The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First Article data.- The first article test plan and test report(s) shall be as required in the contract.

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TABLE I First Article Inspection

Inspections		Rqr Para	Test Para
1.	Inspection covered by subsidiary documents	-	4.4
2.	Interchangeability	3.22	4.18
3.	Mounting Insert	3.11	4.15.1
4.	Solderless-terminal-lug-connectors	3.21.4	4.17
5.	Humidity	3.10	4.14
6.	Visual and Mechanical	3.23	4.19
7.	Power systems and components installation	3.3.1	4.7.1
8.	Power distributional and operational function	3.3.2	4.7.2
9.	Signal (audio) system wiring and components installation	3.3.3	4.7.3
10.	System operation	3.4	4.8
11.	Acoustical noise	3.12	4.16
12.	Vehicular transportation	3.5	4.9
13.	Rail transportation	3.6	4.10
14.	Flat drop	3.7	4.11.1
15.	Rotational drop	3.7	4.11.2
16.	Watertightness	3.8	4.12
17.	Fording	3.9	4.13
18.	Repeat 7. Power systems and components installation	3.3.1	4.7.1
19.	Repeat 8. Power distributional and operational function	3.3.2	4.7.2
20.	Repeat 9. Signal (audio) system wiring & components installation	3.3.3	4.7.3
21.	Repeat 10. System operation	3.4	4.8

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract.

Item	Where Required
Cable assemblies	3.3.4
Suppression, RFI	3.3.5
Finish	3.14
Marking	3.15
Welding	3.29

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection.- Each unit on contract shall be inspected for conformance to the inspections specified in Table II. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

TABLE II - Group A Inspection

Inspection	Rqr Para	Insp Para	AQL % Def Major	Minor
Visual and Mechanical	3.23	4.19	1.0	4.0
Electrical installation				**

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TABLE II - Group A Inspection
(contd)

Inspection	Rqr Para	Insp Para	AQL % Def Major	Minor
Power systems and component installation				
Continuity of conductors	3.3.1.1	4.7.1.1	1.0 for the entire group of power and signal wiring assemblies	
Dielectric breakdown voltage	3.3.1.2	4.7.1.2		
Insulation resistance	3.3.1.3	4.7.1.3		
Power distributional and operational function	3.3.2	4.7.2 thru 4.7.2.1		
Signal (audio) system wiring and components installation				
Continuity of conductors	3.3.3.1	4.7.3.1		
Dielectric and insulation resistance	3.3.3.2	4.7.3.2		

** All defects are major

4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in the order listed in Table II.

4.5.2 Group B inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, except for Interchangeability and Mounting Insert Inspections which are performed prior to Group A. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

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TABLE III - Group B Inspection

Inspection	Rqr Para	Insp Para	AQL % Def (See 4.5.2.1)
Interchangeability	3.22	4.18	4.0
Mounting insert	3.11	4.15.2	4.0
Watertightness	3.8	4.12	2.5
Fording	3.9	4.13	2.5

NOTE: The order of inspection of Group B with respect to Groups A and C shall conform to the order specified in 4.3. The order of inspection within Group B shall be as in 4.5.2.2.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table III shall be as follows:

AQL	Inspection level
4.0%	S-4
2.5%	S-4

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed on the same equipment in the order in which they appear.

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV. Samples shall be selected in accordance with 4.5.3.1 thru 4.5.3.4.

TABLE IV - Group C Inspection

Inspection	Rqr Para	Insp Para
Construction verification	3.2	4.6
Solderless-terminal-lug connection	3.21.4	4.17

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* 6.1 Intended use. The core material covered by this specification is

TABLE IV - Group C Inspection (contd)

Inspection	Rqr Para	Insp Para
Mounting inserts	3.11	4.15.1
System operation	3.4	4.8
Acoustical noise	3.12	4.16
Vehicular Transportation	3.5	4.9
Rail Transportation	3.6	4.10
Drops, Flat and Rotational	3.7	4.11
Humidity	3.10	4.14

NOTE: The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.3. The order of inspection within Group C for all inspections listed shall be as listed in Table IV. Units subjected to Group C Drop Tests shall subsequently be subjected to and meet the watertightness and fording tests for 4.12 and 4.13.

4.5.3.1 Sampling for inspection of construction evaluation.- The Equipment Verification Review, in accordance with paragraph 4.6, shall be performed once on each part and assembly, randomly selected, and on their installation into one of first 10 shelter facilities produced. This inspection shall be performed as the selected unit is being assembled.

4.5.3.2 Sampling for inspection of solderless-terminal-lug-connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a one foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimpling the terminal lugs in production.

4.5.3.3 Sampling for inspection of drop tests.- One sample, representative of the first 10 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the vehicular and rail transportation tests of 4.5.3.4. Thereafter, one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

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4.5.3.4 Sampling for other inspection.- For inspection tests listed below, one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the government.

a. Mounting inserts

b. System operation

c. Acoustical noise

d. Truck transportability

e. Railroad transportability

f. Humidity

4.5.3.5 Group C failures.- Actions required relative to Group C failures shall be as specified in the contract.

4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage. (See 6.4)

4.5.5 Alternate Group A audit and Group B inspection procedures.- The contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plans from MIL-STD-105. If this alternate is selected the procedure of implementation must be in accordance with continuous sampling multilevel plans described in MIL-STD-1235. Sampling plans for the AQL's listed in Table II Group A inspection, and Table III Group B inspection of this specification shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=1/2. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting production contractor shall notify the contracting officer in writing which method he intends to use. The method may not be changed during production without written approval of the government.

4.6 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (See 3.2) cited in the Technical Data Package (as supplemented by all approved engineering changes). The EVR shall consist of the following:

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a. An audit to establish that "as-built" shelter facilities, including individual parts, their use in subassemblies, and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

b. An audit to establish that "as-built" shelter facilities, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, components, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of parts, components, shelter facilities and details proving compliance with 4.6a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the government technical organization cited in the contract. The results of the EVR shall be subject to verification by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government Verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to government personnel during the verification. The contractor shall provide the following:

(1) Segregation of parts, components of shelter facilities, to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

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(3) Adequate administrative support for the EVR.

4.7 Electrical installation tests.- The tests of 4.7.1 through 4.7.3 shall be performed on each shelter facility. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to be supplied shall be 120/208 volts AC, 3 phase, 60 hertz.

4.7.1 Power systems and components installation tests.- (See 3.3.1) Prior to conducting power wiring and components tests of 4.7.1.1, 4.7.1.2, and 4.7.1.3, the following shall be accomplished: (Refer to the power schematic drawing SC-D-629721)

a. Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent, and neon (power indicator, power distribution indicators, and correct and incorrect phase indicators) lamps.

(2) All fluorescent lamp starters.

(3) Blowers, heaters, and air conditioner, if applicable.

b. Remove power distribution box cover panel and disconnect both conductors from the voltmeter and frequency meter terminals. Position or isolate disconnected leads so as not to short to each other or to shelter facility ground.

4.7.1.1 Continuity of conductors.- Continuity resistance tests to meet the requirements of 3.3.1.1 shall be performed on each power circuit in accordance with the following paragraphs. A calibrated low resistance reading ohmmeter or equivalent shall be used in performing the test.

a. Place the following controls in the ON position:

(1) Both fluorescent light switches

(2) Incandescent cold start switch

(3) Blower switch

(4) Main circuit breaker

(5) All branch circuit breaker

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b. Place Blackout/Bypass switch to the Bypass position.

c. Measure continuity resistance as follows:

	FROM	TO
(1)	Pin A of J1	Pin A of J2
(2)	Pin A of J1	Pin X of AC power receptacle J16
(3)	Pin A of J1	Black wire terminations at each incandescent and fluorescent lamp
(4)	Pin A of J1	Brass contact of blower power receptacle J17
(5)	Pin A of J1	Brass contact of all test equipment power receptacles J7, J9 thru J13, J15 and J23
(6)	Pin B of J1	Pin B of J2
(7)	Pin B of J1	Pin Y of AC power receptacle J16
(8)	Pin B of J1	Brass contact of power supply receptacle J14
(9)	Pin B of J1	Brass contact of HTR-1 receptacle J6
(10)	Pin C of J1	Pin C of J2
(11)	Pin C of J1	Pin Z of AC power receptacles J16
(12)	Pin C of J1	Brass contact of HTR-2 receptacle J18
(13)	Pin C of J1	Brass contact of all convenience receptacles J8 and J20 and intercom receptacle J19
(14)	Pin N of J1	Pin N of J2
(15)	Pin N of J1	Pin W of AC power receptacle J16
(16)	Pin N of J1	White wire terminations at each incandescent and fluorescent lamp
(17)	Pin N of J1	Silver contacts of all 115 VAC power receptacles
(18)	Ground terminal E1	Ground contact of all power receptacles and pin G of J1 and J2.

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	FROM	TO
(19)	Pin A of J3	Pin A of operator position power receptacle J21
(20)	Pin B of J3	Pin B of operator position power receptacle J21
(21)	Pin C of J3	Pin C of operator position power receptacle J21
(22)	Pin A of J4	Pin A of operator position power receptacle J22
(23)	Pin B of J4	Pin B of operator position power receptacle J22
(24)	Pin C of J4	Pin C of operator position power receptacle J22

4.7.1.2 Dielectric breakdown voltage.- Dielectric breakdown tests to meet the requirements of 3.3.1.2 shall be performed on each power line circuit in accordance with the following paragraphs. A non-destructive DC dielectric tester shall be used in performing the test.

NOTE: Take preventive measures that will preclude personnel from coming into contact with circuits under test during performance of ALL dielectric and insulation resistance tests.

a. Retain all circuit controls listed under 4.7.1.1a in the ON position.

b. Disconnect at switch S1 all lines to the phase indicator circuitry. Isolate disconnected lines so as not to short to each other or to shelter facility ground.

c. Place Blackout/Bypass switch in the Bypass position.

d. Test between Pin A of J1 and all other pins of J1 strapped to shelter facility ground terminal. Adjust tester for an output of 500 volts DC and maintain this potential for a period of 30 seconds.

e. Repeat step d above between the following test points:

(1) Pin B of J1 and all other pins of J1 strapped to shelter facility ground terminal.

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(2) Pin C of J1 and all other pins of J1 strapped to shelter facility ground terminal.

(3) Pin H of J1 and all other pins of J1 strapped to shelter facility ground terminal.

(4) Pin A of J3 and all other pins of J3 strapped to ground.

(5) Pin B of J3 and all other pins of J3 strapped to ground.

(6) Pin C of J3 and all other pins of J3 strapped to ground.

(7) Pin A of J4 and all other pins of J4 strapped to ground.

(8) Pin B of J4 and all other pins of J4 strapped to ground.

(9) Pin C of J4 and all other pins of J4 strapped to ground.

4.7.1.3 Insulation resistance.- Insulation resistance tests shall be made on each power line conductor. (See 3.3.1.3) Test conditions, preparation and points of measurements shall be as given for 4.7.1.2 above. The insulation resistance of all conductors with respect to ground shall not be less than 100 megohms.

4.7.2 Power distributional and operational functions.- Power distribution and operational function tests shall be made subsequent to the tests of 4.7.1 and with all electrical sources and loads connected. All components which are part of and associated with the electrical power system shall be in-place and connected. Prior to conducting the tests of 4.7.2.1 the following shall be accomplished (refer to power schematic drawing SC-D-629721):

a. Secure blower exhaust port cover in the OPEN position.

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b. Open shelter door.

c. Place all circuit breakers (including main) in the OFF position.

d. Provide and place suitable 120 VAC test lamps or equivalent test indicators in each 115 VAC power receptacles.

e. Place all the lighting control switches in the ON position.

f. Place the Blackout/Bypass switch in the Bypass position.

g. Reconnect voltmeter, frequency meter, and circuitry to phase indicator.

4.7.2.1 Test of circuit breakers and switches.-

a. Apply 120/208 three phase, 60 hertz power to Power In connector J1 at the power entrance box. POWER ON indicator lamp shall light. Correct phase indicator lamp shall light.

b. Rotate meter switch S1 to positions L1, L2, and L3 respectively. The voltmeter shall indicate specific line voltage (approximately 120 volts AC) in each instance. The frequency meter shall indicate 60 hertz in each instance.

c. Place main circuit breaker CB1 in the ON position. Turn each branch circuit breaker ON in the sequence given below. The corresponding indicator lamp shall light and the indication shall be as listed below for each breaker.

Circuit Breaker No.	Indication
(1) CB-2 (Air Conditioner)	208 VAC indication with a voltmeter connected across each phase at the air conditioner power receptacle J16; and 120 VAC indication with the voltmeter connected between each phase and neutral.
(2) CB-3A (Lights)	All fluorescent and incandescent lamps shall light

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Circuit Breaker No.	Indication
(3) CB-3B (Blower)	Blower shall operate and exhaust air from the shelter
(4) CB-4A (Power Supply)	Test lamp in power supply power receptacle shall light
(5) CB-4B (HTR-1)	Test lamp in heater #1 power receptacle shall light
(6) CB-5A (HTR-2)	Test lamp in heater #2 power receptacle shall light
(7) CB-5B (Convenience)	Test lamps in all convenience and intercom power receptacles shall light
(8) CB-6A (Equipment)	Test lamps in all equipment power receptacles shall light
(9) CB-6B (TTY)	Test lamp in TTY receptacle shall light

d. With all circuit breakers remaining in the ON position, operate power control switches to OFF in the sequence given below. The indication shall be as listed for each switch.

Switch	Indication
(1) Incandescent cold start	All incandescent lamps shall extinguish
(2) Fluorescent lights #1	Half of roadside and curbside fluorescent lamps shall extinguish
(3) Fluorescent lights #2	Remaining fluorescent lamps shall extinguish
(4) Blower	Blower shall cease to operate

e. Reset all lighting switches to restore lights and perform the following blackout test:

(1) Place Blackout/Bypass switch in the Blackout position. All fluorescent and incandescent lamps shall extinguish.

(2) Close shelter door. All fluorescent and incandescent lamps shall light.

(3) Open shelter door and replace Blackout/Bypass switch to Bypass.

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f. Reset blower switch to ON.

g. Place main circuit breaker OFF. Blower shall cease to operate. All test lamps, fluorescent lamps and incandescent lamps shall extinguish.

4.7.3 Signal (audio) system wiring and components.- (Refer to the signal schematic drawing SC-D-629722)

4.7.3.1 Continuity.- Continuity resistance tests shall be performed on each signal circuit conductor. A low resistance reading ohmmeter or equivalent continuity tester shall be used. (An automatic tester may be used, subject to approval by the government cognizant engineer). Continuity measurements shall be tested between the following points and meet the requirements of 3.3.3.1:

	FROM U-187/G	TO OPER POS CONN	
SYS-1	Pin 1A, 9A	Pin D	Operator Position
	Pin 1B, 9B	Pin C	
	Pin 2A, 10A	Pin B	
	Pin 2B, 10B	Pin A	
	Pin 3A, 11A	Pin M	
	Pin 3B, 11B	Pin L	
	Pin 4A, 12A	Pin K	
	Pin 4B, 12B	Pin J	
	Pin 5A, 13A	Pin H	
	Pin 5B, 13B	Pin G	
	Pin 6A, 14A	Pin F	
	Pin 6B, 14B	Pin E	
	Pin 7A, 15A	Pin S	
	Pin 7B, 15B	Pin R	
	Pin 8A, 16A	Pin P	
	Pin 8B, 16B	Pin N	
SYS-2	Pin 1A, 9A	Pin D	Remote operator Position
	Pin 1B, 9B	Pin C	
	Pin 2A, 10A	Pin B	
	Pin 2B, 10B	Pin A	
	Pin 3A, 11A	Pin M	
	Pin 3B, 11B	Pin L	
	Pin 4A, 12A	Pin K	
	Pin 4B, 12B	Pin J	

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	FROM U-187/G	TO OPER POS CONN	
SYS-2	Pin 5A, 13A	Pin H	Remote operator Position
	Pin 5B, 13B	Pin G	
	Pin 6A, 14A	Pin F	
	Pin 6B, 14B	Pin E	
	Pin 7A, 15A	Pin S	
	Pin 7B, 15B	Pin R	
	Pin 8A, 16A	Pin P	
	Pin 8B, 16B	Pin N	

FROM U-187/G SYS-3	TO BINDING POSTS
Pin 1A	CH-1 2WT/R A
Pin 1B	CH-1 2WT/R B
Pin 2A	CH-1 4WT A
Pin 2B	CH-1 4WT B
Pin 3A	CH-2 2WT/R A
Pin 3B	CH-2 2WT/R B
Pin 4A	CH-2 4WT A
Pin 4B	CH-2 4WT B
Pin 5A	CH-3 2WT/R A
Pin 5B	CH-3 2WT/R B
Pin 6A	CH-3 4WT A
Pin 6B	CH-3 4WT B
Pin 7A	CH-4 2WT/R A
Pin 7B	CH-4 2WT/R B
Pin 8A	CH-4 4WT A
Pin 8B	CH-4 4WT B
Pin 9A	CH-5 2WT/R A
Pin 9B	CH-5 2WT/R B
Pin 10A	CH-5 4WT A
Pin 10B	CH-5 4WT B
Pin 11A	CH-6 2WT/R A
Pin 11B	CH-6 2WT/R B
Pin 12A	CH-6 4WT A
Pin 12B	CH-6 4WT B
Pin 13A	CH-7 2WT/R A
Pin 13B	CH-7 2WT/R B
Pin 14A	CH-7 4WT A
Pin 14B	CH-7 4WT B
Pin 15A	CH-8 2WT/R A
Pin 15B	CH-8 2WT/R B

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FROM U-187/G SYS-3	TO BINDING POSTS
Pin 16A	CH-8 4WT A
Pin 16B	CH-8 4WT B
Pin 17A	CH-9 2WT/R A
Pin 17B	CH-9 2WT/R B
Pin 18A	CH-9 4WT A
Pin 18B	CH-9 4WT B
Pin 19A	TTY REC A
Pin 19B	TTY REC B
Pin 20A	TTY XMT A
Pin 20B	TTY XMT B
Pin 21A	TA-341 #1 REC A
Pin 21B	TA-341 #1 REC B
Pin 22A	TA-341 #1 XMT A
Pin 22B	TA-341 #1 XMT B
Pin 23A	TA-341 #2 REC A
Pin 23B	TA-341 #2 REC B
Pin 24A	TA-341 #2 XMT A
Pin 24B	TA-341 #2 XMT B
Pin 25A	TA-312/PT A
Pin 25B	TA-312/PT B
Pin 26A	INTERCOM A
Pin 26B	INTERCOM B

Inside shelter facility short REC pairs and XMT pairs of lines connecting to the telegraph converter, teletypewriter and four wire telephone sets. Short pair of lines connecting to the TA-312/PT and INTERCOM.

	FROM U-187/G	TO U-187/G
	Pin 19A	Pin 19B
	Pin 20A	Pin 20B
SYS-3	Pin 21A	Pin 21B
	Pin 22A	Pin 22B
	Pin 23A	Pin 23B
	Pin 24A	Pin 24B
All	Pin 25A	Pin 25B
SYS's	Pin 26A	Pin 26B

Following completion of the continuity tests, remove the above mentioned shorts.

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4.7.3.2 Dielectric strength and insulation resistance.- Dielectric strength and insulation resistance tests which meet the requirements of 3.3.3.2 shall be performed on each audio-line circuit in accordance with the following paragraphs. A non-destructive dielectric and insulation resistance test instrument and suitable test jigs shall be used in performing the test. (An automatic tester may be used, subject to approval by the government cognizant engineer).

a. Using appropriate test jigs connect all pins of Signal 1 connector to shelter facility ground terminal.

b. Connect tester between shelter facility ground terminal and Pin 1A of Signal 1 connector. Disconnect Pin 1A from ground.

c. Adjust tester to 500 volts DC output and maintain this potential for a period of 5 seconds. The requirements of 3.3.3.2 shall be met. Re-establish ground connection to the pin under test.

d. Repeat step c above for each of the remaining pin terminations of Signal 1 connector; each time ungrounding the pin being tested and reconnecting to ground following completion of test.

e. Repeat steps a thru d above for Signal connectors 2 and 3.

4.8 Systems operation tests.- Install all operational and test equipments in their respective locations and connect all power and signal cable assemblies to their designated receptacles and connectors. (See 3.4) (Refer to drawing SC-D-629640 for layout of equipments)

4.8.1 Test of operators position and remote operators position.- Unless otherwise specified, the following test procedures applies to the operators position and remote operators position.

a. Interconnect Signal 1 and Signal 2 connectors by the use of Cable Assembly CX-4566/U (25 Ft).

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b. Connect operators position and remote operators position to their designated internal power receptables.

c. Perform the following POWER ON observations:

(1) Set PWR/LOGIC switch to ON.

(2) Depress RESET switch.

(3) COMMON RELEASE lamp should be on. All other lamps should be off.

d. Perform the following lamp test:

(1) Depress LAMP TEST switch.

(2) All lamps shall go on except RERING, CLEAR/WIPEOUT and COMMON RELEASE lamps.

(3) Turn LAMP INTENSITY control from LO to HI. Intensity of all lamps shall vary from full off to full on. Adjust LAMP INTENSITY control for a comfortable level of illumination.

e. Perform the following keyset pushbuttons check:

(1) Depress 1 pushbutton and listen for tone in earphone.

(2) Repeat for the remaining keyset pushbuttons.

f. Perform the following pushbutton switches and lamps check:

(1) Depress CALL/SEIZE switch and listen for tones in earphone. After one second delay, the CALL/SEIZE lamp and QUEUE ADVANCE lamp shall go on and the COMMON RELEASE lamp shall extinguish. Listen for dial tone.

(2) Depress COMMON RELEASE switch and listen for tones in earphone. CALL/SEIZE and QUEUE ADVANCE lamps shall extinguish and COMMON RELEASE lamp shall go on.

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(3) Depress RELEASE 1 switch and listen for tones in earphone. RELEASE 1 lamp shall light for approximately two seconds.

(4) Depress RELEASE 2 switch and listen for tones in earphone. RELEASE 2 lamp shall light for approximately two seconds.

(5) Repeat step f(1) above.

(6) Depress QUEUE ADVANCE switch and listen for tones in earphone. COMMON RELEASE lamp shall go on and QUEUE ADVANCE and CALL/SEIZE lamps shall extinguish if there is no call in queue. If call is in queue, after one second the COMMON RELEASE lamp extinguishes and QUEUE ADVANCE and CALL/SEIZE lamps remain on.

(7) Depress HOLD switch and listen for tones in earphone.

(8) Depress RESET switch.

(9) Depress CLEAR/WIPEOUT switch and listen for tones in earphone.

(10) Depress RERING switch and listen for tones in earphone.

(11) Set MODE switch to UNSTAFFED position and listen for tones in earphone. Return MODE switch to STAFFED position.

(12) Depress INTERCOM switch and listen for tone burst in earphone.

(13) Depress INTERCOM switch again. INTERCOM lamp shall extinguish.

g. Perform the following VOICE circuits tests on the operators positon.

(1) Call remote operator position from operator position.

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(2) When remote operator position answers, place call on hold by depressing operator's position HOLD switch. HOLD lamp shall go on. COMMON RELEASE lamp shall go on and CALL/SEIZE and QUEUE ADVANCE lamps shall extinguish.

(3) Retrieve call from hold by depressing operator's position HOLD switch again. HOLD lamp shall extinguish. CALL/SEIZE and QUEUE ADVANCE lamps shall go on and COMMON RELEASE lamp shall extinguish.

(4) Establish two way voice conversation.

(5) Release remote operators position by depressing RELEASE 1 switch at the operators position. RELEASE 1 lamp shall go on for one second, then extinguish.

h. Perform the following incoming calls to the operators position:

(1) At the remote operators position establish a precedent call, and on-hook before operator answers.

(2) The STATUS QUEUE lamp flashes at the operators position to indicate an incoming precedence call.

(3) STATUS QUEUE lamp extinguishes, to indicate that the incoming call from the remote operators position has gone on-hook before being answered, or another operator has answered.

(4) At the remote operators position establish a routine call.

(5) The STATUS QUEUE lamp goes on steady at the operators position to indicate an incoming routine call.

(6) Answer call at operators position by depressing QUEUE ADVANCE switch. Establish two way voice conversation.

(7) Release call at operators position by depressing COMMON RELEASE switch. COMMON RELEASE lamp shall go on and QUEUE ADVANCE and CALL/SEIZE lamp shall extinguish.

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i. Repeat voice circuits tests and incoming calls of steps g and h above for the remote operators position.

j. Perform the following intercom circuit tests:

(1) Depress INTERCOM switch at the operators position. INTERCOM lamp shall go on at the operators position. INTERCOM lamp shall flash at the remote operators position.

(2) Depress INTERCOM switch at the remote operators position. INTERCOM lamp at the remote operators position shall go on steady. Establish two way voice communication on the intercom circuit.

(3) At both positions release call by depressing INTERCOM switch. INTERCOM lamp shall extinguish at both positions.

(4) Depress INTERCOM switch at the remote operators position. INTERCOM lamp shall go on at the remote operators position. INTERCOM lamp shall flash at the operators position.

(5) Depress INTERCOM switch at the operators position. INTERCOM lamp at the operators position shall go on steady. Establish two way voice communication on the intercom circuit.

(6) At both positions release call by depressing INTERCOM switch. INTERCOM lamp shall extinguish at both positions.

k. Verified that the STATUS ALARM lamp on each position goes on when an alarm condition is simulated, and that the lamp extinguishes when the alarm is reset.

4.8.2 Test of teletypewriter circuits.-

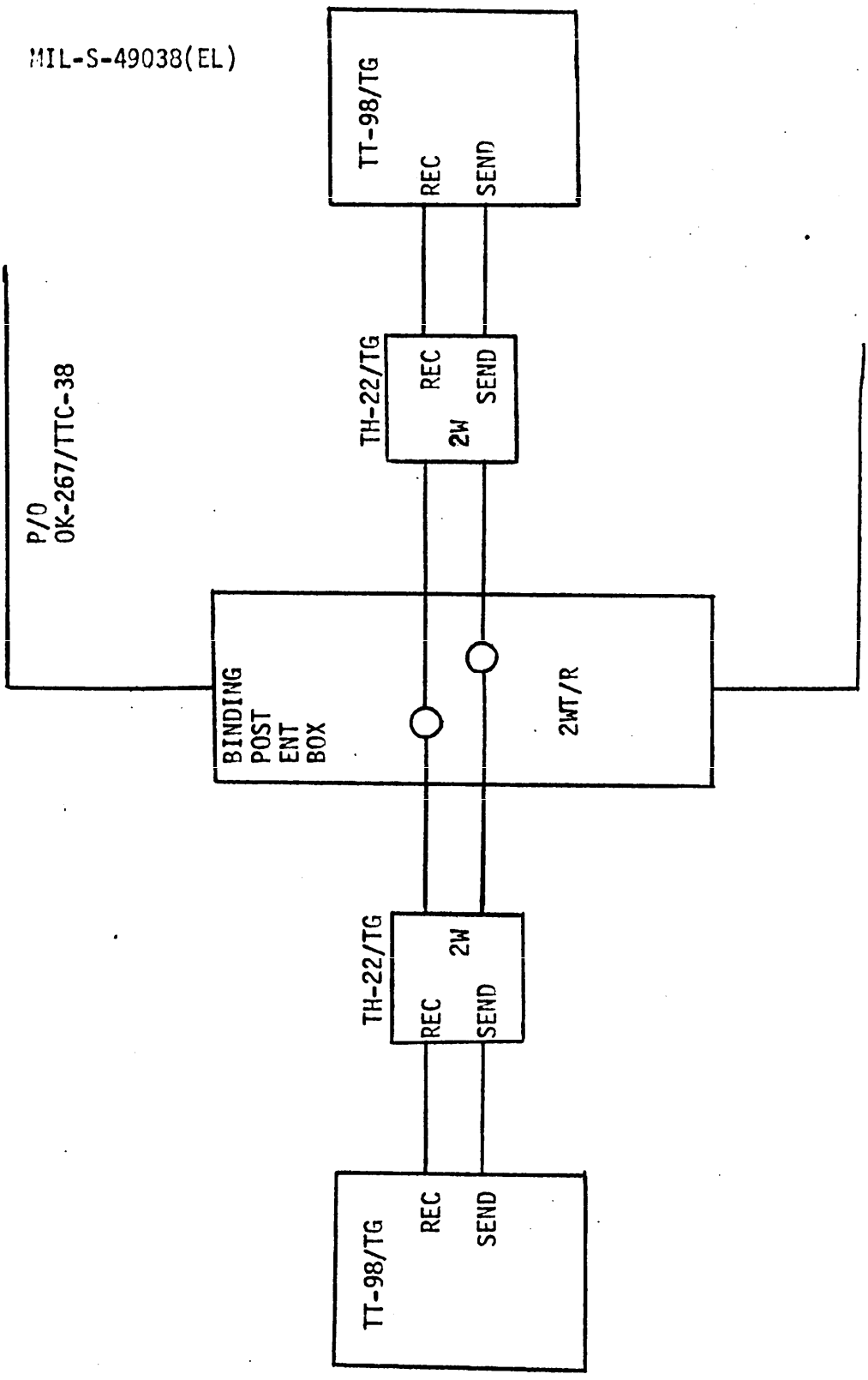
a. Place an external Teletypewriter TT-99/TG and Telegraph Converter TH-22/TG across the TTY binding posts as shown in Figure 1.

b. Set TH-22/TG NORM/REC/SEND control to NORM; 4W-2W-TEL switch to 2W; VF-20 switch to VF and MODE ONE/MODE TWO switch to MODE ONE.

c. Arrange teletypewriters for half duplex operations.

d. Send a series of "FOX" transmission messages on each teletypewriter (not simultaneously). The transmitted messages should be printed at the receiving teletypewriter, including the sending teletypewriter. No errors shall exist in the reception of the messages.

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TELETYPEWRITERS HALF DUPLEX OPERATION

FIGURE 1

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4.8.3 Check of DC power supply.-

a. Place a DC voltmeter (0-50 volt range) across the DC terminals located on the roadside raceway.

b. Turn power supply on. Under no load conditions vary power supply voltage control to 50 volts. DC voltmeter across the DC terminals shall read 50 volts DC.

4.9 Vehicular transportation test.- The facility shall be loaded to simulate the SK-267( )(V)/TTC-38(V) using live and dummy loads (See 3.2b). The exhaust blower and air conditioner shall be mounted in their normal operating positions. Using the lift and tie down slings furnished with the bare shelter, the facility shall be secured as indicated on the shelter transportation data plate, to the bed of a government furnished 2-1/2 ton, M-35 truck. The tire pressure shall be set for cross-country travel as listed on the truck dashboard. The facility shall be centralized and blocked to prevent shifting. The facility shall be transported over the specially prepared course described in 4.9.1 for 10 laps at each speed of 5, 10, 15 and 18-20 MPH. A lap is defined as traversing the course in one direction. Upon completion of this test, the facility shall not have sustained any of the types of damage cited in 3.5.

4.9.1 Vehicular transportation course.- The course of the military truck transportation test shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1-inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. Each timber shall be anchored down securely by any effective means. The ground on which the timbers are placed shall be hard and level.

4.10 Rail transport, impact test.- The shelter facility loaded as specified in paragraph 4.9, secured to a flatcar (the test flatcar) and coupled to two other flatcars shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter facility tie down slings (contractor furnished), shall be as specified in Figure 2 and shall meet the provisions of the test as specified in paragraph 4.10.1. The test flatcar

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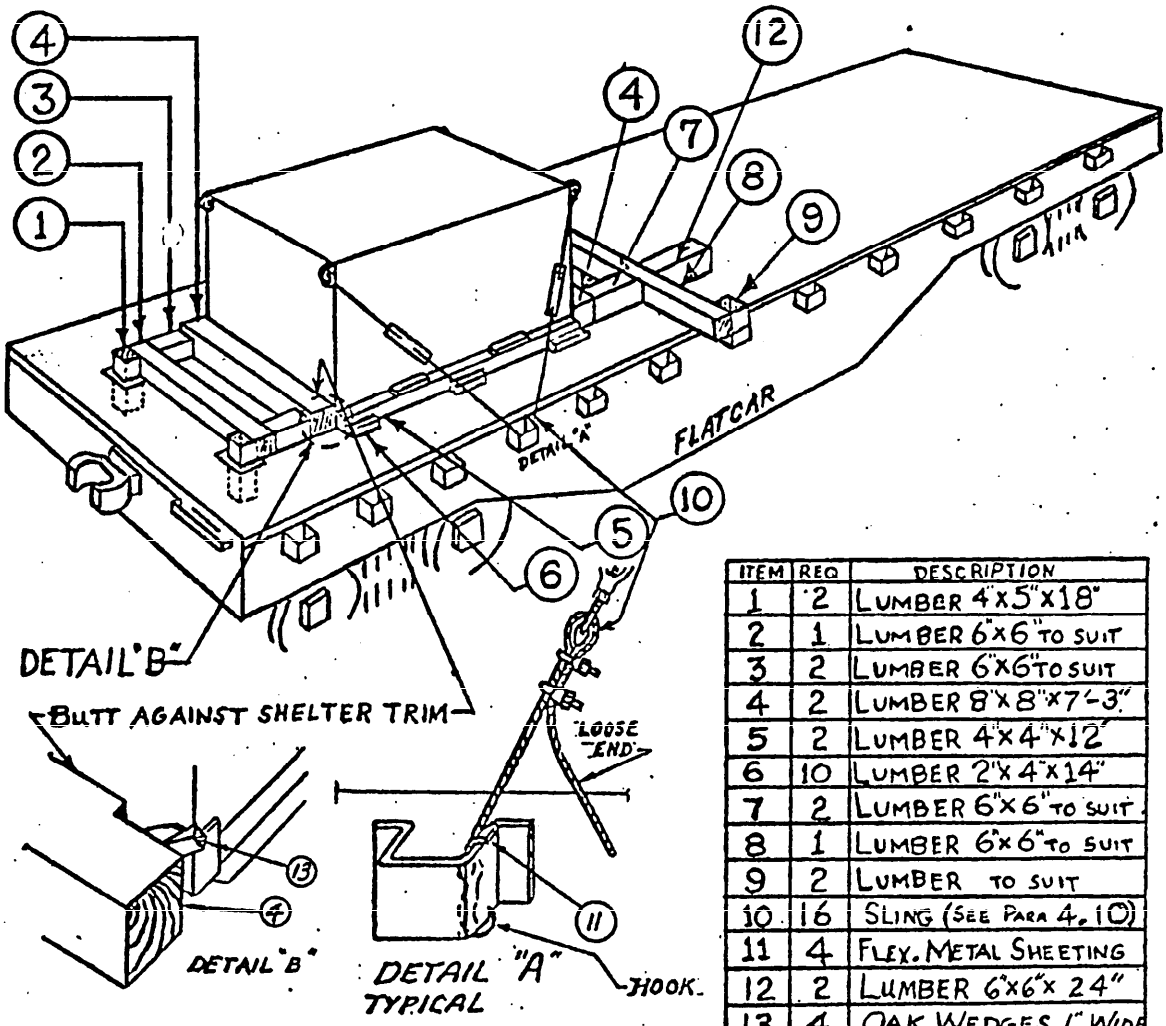
(less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in paragraph 4.10.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds, traveling at 9 miles per hour (or equivalent energy conditions acceptable to the government technical representative), shall be impacted against the stationary loaded test car coupled to the two empty flatcars, all with brakes off. All cars shall have standard (rigid) draft gear. The test shall consist of four impacts, two from each end of the shelter facility. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter facility shall meet the requirements of paragraph 3.6.

4.10.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall be notched or shimmed to contact the trim along the four sides of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred above. All other blocking shall be firmly secured as shown in the figure. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter 6 x 19, flexible steel, preformed wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9,000 pounds minimum. Before each impact, each sling shall be taut (600 - 800 pounds tension).

4.10.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, crushed concrete or metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

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ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"X5"X18"
2	1	LUMBER 6"X6" TO SUIT
3	2	LUMBER 6"X6" TO SUIT
4	2	LUMBER 8"X8"X7'-3"
5	2	LUMBER 4"X4"X12'
6	10	LUMBER 2"X4"X14"
7	2	LUMBER 6"X6" TO SUIT
8	1	LUMBER 6"X6" TO SUIT
9	2	LUMBER TO SUIT
10	16	SLING (SEE PARA 4.10)
11	4	FLEX. METAL SHEETING
12	2	LUMBER 6"X6"X 24"
13	4	OAK WEDGES 1" WIDE

NOTES:
a. The blocking lumber shall be notched to accomodate towing eyes, and bolts. Wedge item 13 between towing corner forging and item 4 notch (Detail B)
b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

FIGURE 2 - SHELTER SECURED ON FLATCAR

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b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

4.11 Drop tests.-

4.11.1 Flat drop.- The shelter facility loaded as in 4.9, shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.7.

4.11.2 Rotational drop.- The shelter facility loaded as in 4.9 shall have one skid resting on a nominal 4 inch high timber. The outer edge of the opposite skid shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.7.

4.12 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc., shall be subjected to the following two part test: (NOTE: Perform test with inside cover of entrance boxes removed).

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Company, Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches.

All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See Figure 3 for nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.13, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door-end panel). Upon

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completion of the above test, the shelter facility shall meet the requirements of 3.8. To determine that no water has entered the shelter panels, a minimum of four (4) 1/8 inch diameter holes (one at each location specified below) shall be drilled through the exterior surface of the shelter floor at the following locations:

(1) Front end, 2 1/2 inches inboard from roadside skid, 1 inch inboard from front face of trim angle.

(2) Front end, 2 1/2 inches inboard from curbside skid, 1 inch inboard from front face of trim angle.

(3) Door end, 2 1/2 inches inboard from roadside skid, 1 inch inboard from front face of trim angle.

(4) Door end, 2 1/2 inches inboard from curbside skid, 1 inch inboard from front face of trim angle.

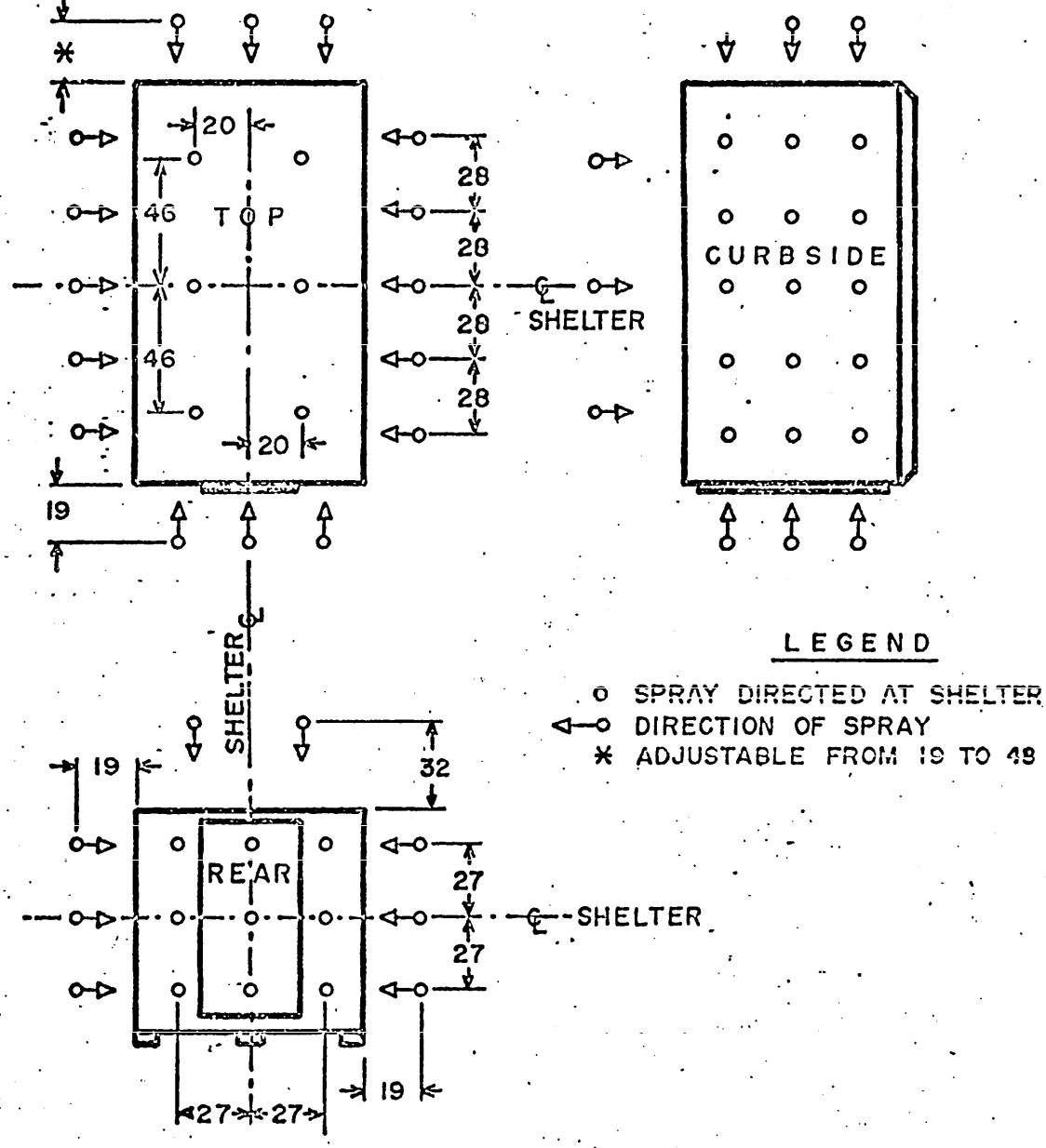
After this determination, the holes shall be sealed with rivets and sealer.

(NOTE: For First Article and Group C tests only, the air conditioner, with brackets, and the exhaust blower and housing with exhaust vent closed shall be mounted outside the shelter in the operating position. For Group B tests the air conditioner opening shall be secured with the designated blank off cover and the blower and housing secured in the storage position and the blower opening secured with the exhaust blower cover.

b. With the air conditioner and bracketry removed, the opening secured with the air conditioner blank off cover, and the exhaust blower and housing in its storage position, the blower opening secured with the exhaust blower cover, and all other covers opened to their operating position (except the shelter door vent panel), repeat a above.

4.13 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 21 inches (measured from the bottom of the skids) and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tie downs. To determine that no water has entered the shelter panels, a minimum of four (4) 1/8 inch diameter holes shall be drilled adjacent to those previously drilled after the watertightness test of paragraph 4.12. After this determination, the holes shall be sealed with rivets and sealer.

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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 3

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4.14 Humidity test.- A hardware sample of five of each size and type (if the total quantity of any size or type is less than five, all shall be tested) of bolt, screw, nut, flat washers, lock washers and rivnuts shall be randomly selected for test. The hardware sample shall be subject to the humidity test specified in Method 507, Procedure II of MIL-STD-810 to determine conformance to the requirements of paragraph 3.10.

4.15 Mounting insert tests.-

4.15.1 First Article and Group C insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

4.15.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE V - Mounting Inserts Loads

Insert Size	Preproduction Test		Production Test	
	Axial(lbs)	Torque(in lbs)	Axial(lbs)	Torque(in lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

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4.16 Acoustical noise test.- The shelter facility shall be located in an area remote from excessive machinery, vehicle and other noise generating sources to reduce ambient noise to a minimum, with the shelter door closed. Only necessary test personnel shall be present in the shelter. Using Octave Band Analyzer, General Radio Corporation, Model 1558A, or equal, and Microphone, Piezoelectric, General Radio Corporation, Model 1560-P6 or equal, record ambient noise levels over the frequency ranges specified in 3.12 at the positions specified in paragraph 4.15.1. Operate the air conditioner in the cooling, high fan speed mode, and repeat noise level recordings at the same frequency levels and positions as above. Repeat above procedure with the exhaust blower operating and the air conditioner off. The noise levels, corrected from ambient shall meet the requirements of 3.12.

4.16.1 Measurement location.- Noise measurements shall be made at the following locations:

a. Remote operators position, 71" from front wall, 20" from roadside wall, 49" above floor.

b. Supervisors position, 39" from front wall, 20" from curbside wall, 49" above floor.

c. Maintenance position, 13" from front wall, 20" from roadside wall, 49" above floor.

4.17 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lug shall be tested for conformance to 3.21.4.

4.18 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.22. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

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b. Size and mounting dimensions of racks and cabinets.

4.19 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.23 and Table VI.

TABLE VI - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated. 2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible. 3. Welds - cracked or porous. Inspection of welded assemblies, such as equipment racks, entrance boxes, etc., should be made prior to final paint application. 4. Sealer improperly applied (holes separations, or lack of adhesion). 5. Mounting inserts missing or mislocated. 6. Burrs or sharp edges in wire duct not removed. 7. Equipments or bracketry loose or missing. 8. Doors, etc., inoperative. 9. Electrical receptacles, switches, jacks, connectors, wire ducts or other electrical components loose, improperly located or inoperative. 10. Locking or holding devices missing or inoperative. 11. Gaps between sections of wire duct exceed 1/16". 12. The blower housing does not align properly with the mounting inserts of the shelter facility. 13. The air conditioner cover does not align properly with the mounting inserts of the shelter facility. 14. Ground stud not properly sealed. 15. Bulkhead Power and Signal connector fasteners not sufficiently tight. 16. Electrical schematics missing.

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TABLE VI - Classification of Visual and Mechanical Defects (contd)

Classification	Defects
Minor	1. Doors not easily operable. 2. Fastening devices difficult to operate. 3. Finish - abrasions or scratches. 4. Burrs and sharp edges on bracket not removed. 5. Screws loose. 6. Gaskets, not completely and securely adhered. 7. Electrical schematics not properly adhered to wall.

4.20 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.21 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.7 Power and Signal Wiring

Paragraph 4.19 Visual and Mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the shelter facility shall be in accordance with MIL-P-55505.

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6.1 Intended use.- Shelter, Electrical Equipment S-541( )/TTC-38(V) is the shelter facility portion of the OK-267( )(V)/TTC-38(V). As the OK-267( )(V)/TTC-38(V), it is used with the Central Office, Telephone, Automatic AN/TTC-38. It houses operator positions for manual control of the Automatic Central Office and provides equipment, facilities for testing, storage, and maintaining of operational components of the Automatic Central Office. The following operational equipments and a portion of the major test equipments forming the Control-Test-Maintenance assemblage OK-267( )(V)/TTC-38(V) are listed below:

a. 1 each Air Conditioner, Horizontal Compact, 9000 BTU.

b. 1 each C-9434/TTC-38(V) Control, Remote, Switchboard.

c. 1 each TS-421( )/U Audio Oscillator

d. 1 each TT-98/FG Teletypewriter

e. 1 each LS-147( )/FI Intercom

f. 1 each TA-341/PT Telephone

g. 1 each TA-312/PT Telephone

h. 1 each AN/USM-207 Frequency Counter

i. 1 each AN/USM-296 Oscilloscope

j. 1 each Resistive Load Bank

k. 1 each DC Power Supply

l. 1 each Pulse Generator

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

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b. First Article and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads or the government furnished live loads (operational and test equipment) as specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-541( )/TTC-38(V).

d. A list of equipments to be government loaned for testing purposes.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.5)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-541R/TTC-38(V). The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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6.7 Tropicalization of material.- All materials of construction are specified on the drawings and are Fungus-inert. Substitution of any materials other than specified must be approved by the contracting officer. The material will only be approved after the contractor tests the material for Fungus-Resistance satisfactorily in accordance with requirement 508 of MIL-STD-810.

Custodian:
Army - EL

Preparing Activity:
Army - EL

Project No. 5410-A174

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-49038(EL) SHELTER, ELECTRICAL EQUIPMENT S-541( )/TTC-38(V)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

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